

INTERNSHIP PROJECT REPORT TASK-1

Title

Red Team | Blue Team | Purple Team: Solve the HawkEye lab and analyze the lab and create 3 reports on it. Report 1: Red Team Report[how red team will work on this lab], Report 2: Blue Team Report[how blue team will work on this lab], Report 3: Purple Team Report[create a handover report on finding, both on attack and defense tactics used to solve the lab].

1. Introduction

As part of my cybersecurity internship, I was assigned the task of performing a complete analysis of the HawkEye malware investigation from the perspectives of the Red Team, Blue Team and Purple Team. The objective of this project was to understand how an attacker steals sensitive information, how defenders detect and investigate the attack and how both teams work together to improve an organization's security.

For this project, I used the HawkEye Lab available on the CyberDefenders platform. The provided PCAP file was analyzed using Wireshark in Kali Linux. During the investigation, I examined different network protocols including DNS, HTTP, TCP and SMTP to identify malicious activities, suspicious communications, downloaded malware, email transmissions and stolen credentials.

2. Background

Cyberattacks are becoming more common, and many attackers use malware to steal sensitive information such as usernames, passwords and personal data. To investigate these attacks, cybersecurity professionals analyze captured network traffic to understand how the attack happened and what information was affected.

For this task, I downloaded the HawkEye Lab from the CyberDefenders platform and opened the provided PCAP file in Wireshark using Kali Linux. During the analysis, I examined different types of network traffic, including HTTP and SMTP, to identify suspicious activities. I also observed the malware download, followed network communication, decoded the captured credentials and understood how sensitive information was transmitted.

3. Learning Objectives

The main objective of this project was to understand how malware-related network traffic can be investigated using Wireshark and how different cybersecurity teams contribute during a security incident.

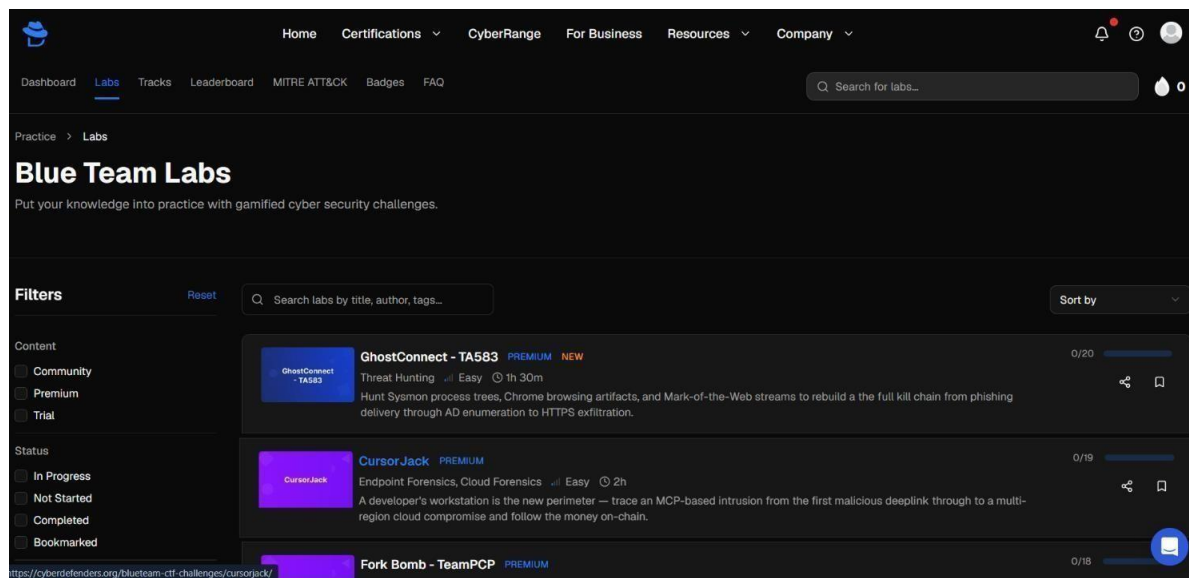
The main objectives of this project were:

- To understand the basics of network traffic analysis and digital forensics.
- To learn how to analyze PCAP files using Wireshark.
- To identify suspicious HTTP, TCP, DNS and SMTP network traffic.
- To investigate malware communication and understand how data is transmitted.
- To understand the roles and responsibilities of the Red Team, Blue Team, and Purple Team.

4. Activities and Tasks Performed

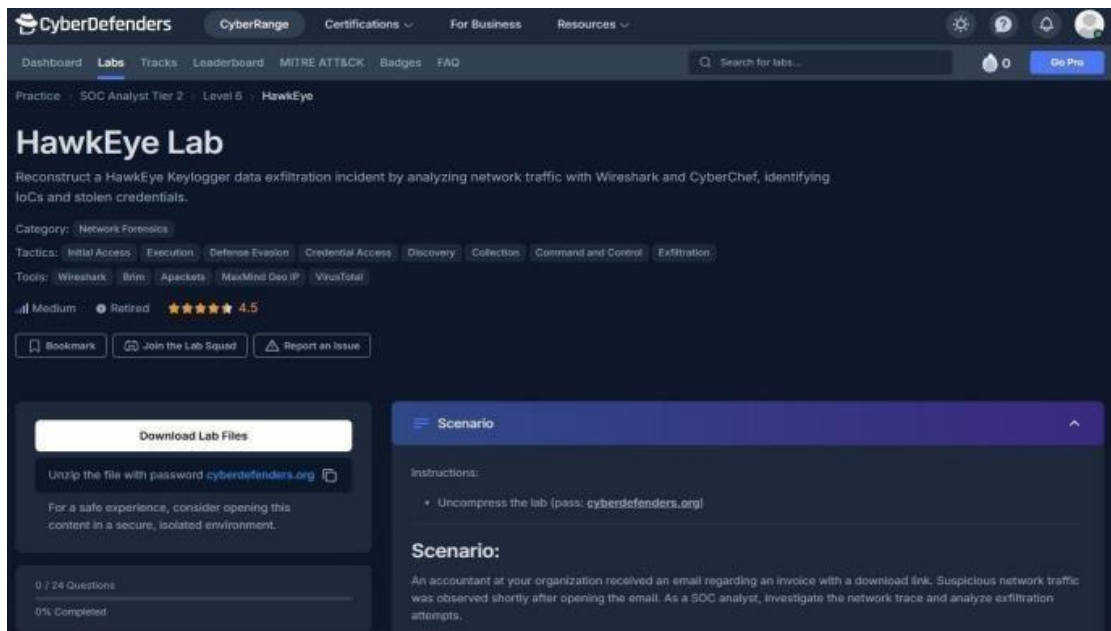
Step 1 - Creating a CyberDefenders Account

I began the project by visiting the CyberDefenders platform and creating a personal account. After completing the registration process, I logged in to the platform and explored the Blue Team Labs section. This gave me access to various cybersecurity challenges designed to improve practical investigation skills.



Step 2 - Accessing the HawkEye Lab

After successfully logging into CyberDefenders, I searched for the HawkEye Lab in the Blue Team Labs section and opened the challenge page. I carefully reviewed the lab description, scenario, objectives and the tools recommended for the investigation. The scenario explained that suspicious network activity was detected after an employee opened a malicious email attachment and my task was to investigate the incident.



The screenshot displays the CyberDefenders web interface for the HawkEye Lab. The top navigation bar includes links for Dashboard, Labs, Tracks, Leaderboard, MITRE ATT&CK, Badges, and FAQ. A search bar is present with the text "Search for labs...". The main content area is titled "HawkEye Lab" and describes the task: "Reconstruct a HawkEye Keylogger data exfiltration incident by analyzing network traffic with Wireshark and CyberChef, identifying IoCs and stolen credentials." The category is "Network Forensics". Tactics listed include Initial Access, Execution, Defense Evasion, Credential Access, Discovery, Collection, Command and Control, and Exfiltration. Tools listed include Wireshark, Bin, Apeckets, MaxMind Geo IP, and VirusTotal. The lab is rated "Medium" difficulty, "Refined", and has a 4.5 star rating. Below the lab details, there is a "Download Lab Files" button and a "Scenario" section. The scenario text reads: "An accountant at your organization received an email regarding an invoice with a download link. Suspicious network traffic was observed shortly after opening the email. As a SOC analyst, investigate the network trace and analyze exfiltration attempts."

CyberDefenders

CyberRange Certifications For Business Resources

Dashboard Labs Tracks Leaderboard MITRE ATT&CK Badges FAQ

Search for labs...

Practice SOC Analyst Tier 2 Level 6 HawkEye

HawkEye Lab

Reconstruct a HawkEye Keylogger data exfiltration incident by analyzing network traffic with Wireshark and CyberChef, identifying IoCs and stolen credentials.

Category: Network Forensics

Tactics: Initial Access Execution Defense Evasion Credential Access Discovery Collection Command and Control Exfiltration

Tools: Wireshark Bin Apeckets MaxMind Geo IP VirusTotal

Medium Refined 4.5

Bookmark Join the Lab Squad Report an Issue

Download Lab Files

Unzip the file with password `cyberdefenders.org`

For a safe experience, consider opening this content in a secure, isolated environment.

0 / 24 Questions
0% Completed

Scenario

Instructions:

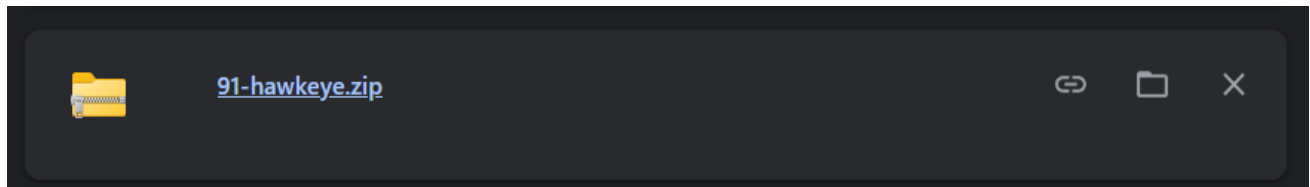
- Uncompress the lab (pass: `cyberdefenders.org`)

Scenario:

An accountant at your organization received an email regarding an invoice with a download link. Suspicious network traffic was observed shortly after opening the email. As a SOC analyst, investigate the network trace and analyze exfiltration attempts.

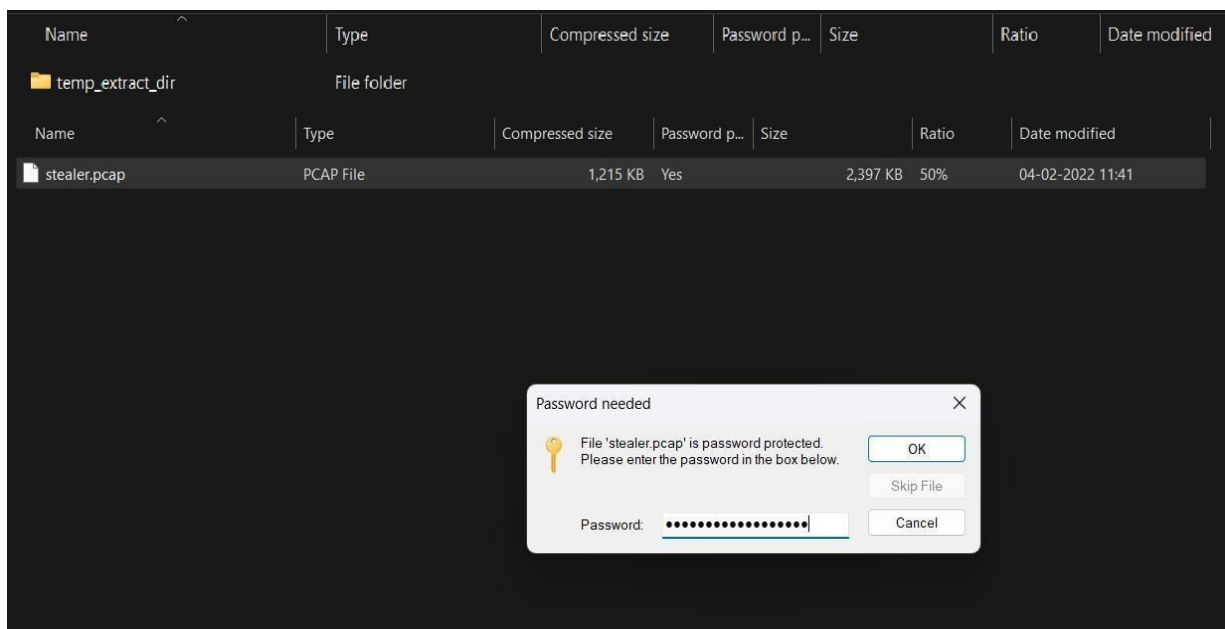
Step 3 - Downloading the Lab File

After opening the HawkEye lab, I downloaded the provided 91-hawkeye.zip file. This file contained the files needed for the investigation. I saved it in my Kali Linux system so I could extract and analyze the lab data in the next steps.



Step 4 - Extracting the Lab File

After downloading 91-hawkeye.zip, I opened the archive and selected the extraction option. The archive contained a folder named temp_extract_dir. I then continued the extraction process and entered the password provided with the lab. After entering the correct password, the protected file stealer.pcap was extracted successfully.



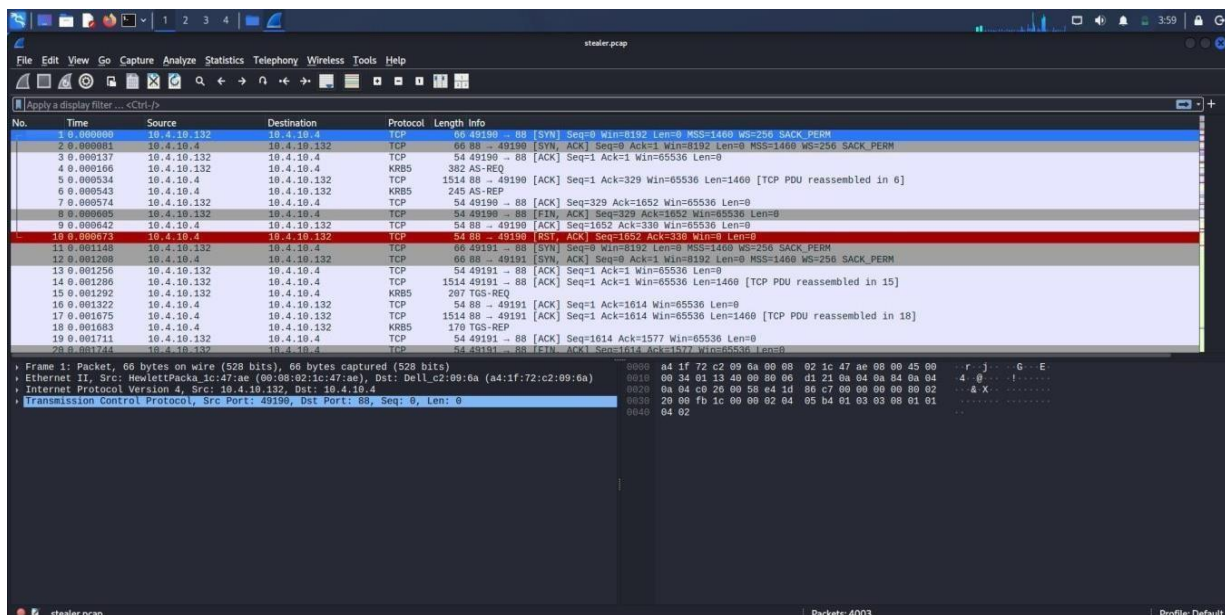
Step 5 - Preparing the Investigation Environment

Before starting the analysis, I launched my Kali Linux virtual machine using Oracle VirtualBox. I used Kali Linux because it provided the tools needed for network traffic analysis and digital forensic investigation.



Step 6 - Opening the PCAP File

I opened the **stealer.pcap** file in Wireshark. The packet list was displayed with different network protocols and communication details. This allowed me to start examining the captured network traffic for suspicious activity.

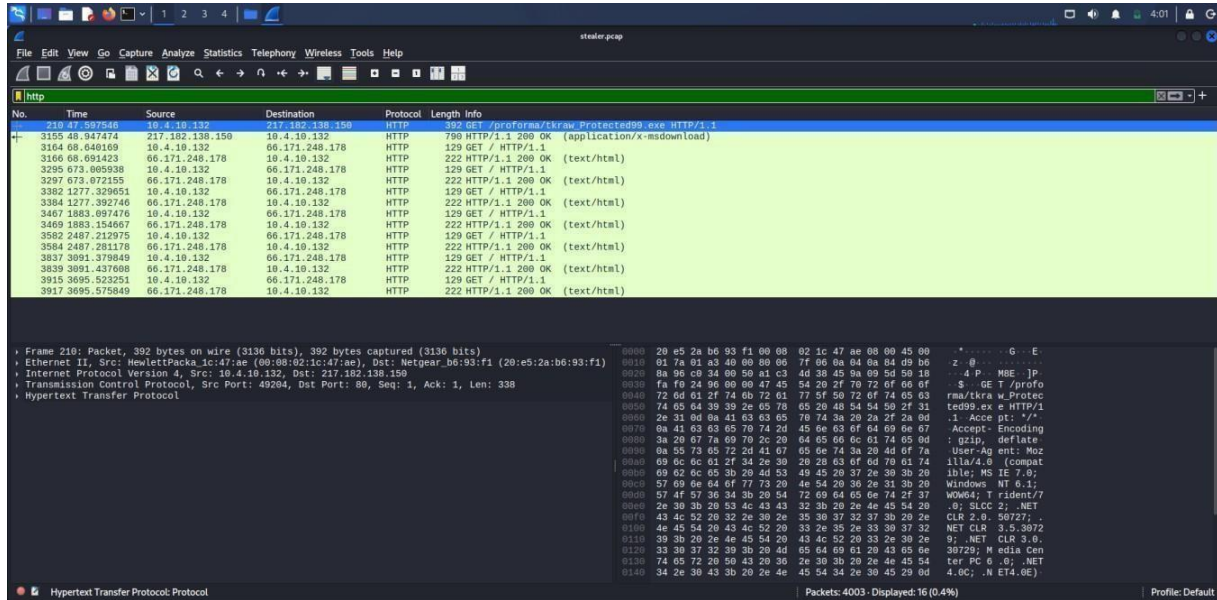


Step 7 - Analyzing HTTP Traffic

To check the web traffic in the PCAP file, I filtered the network traffic to display only HTTP packets.

In the Wireshark filter bar, I entered: **http**

After applying the filter, I examined the HTTP requests and responses. During the analysis, I found a GET request for a file named **tkraw_Protected99.exe** from the server 217.182.138.150.

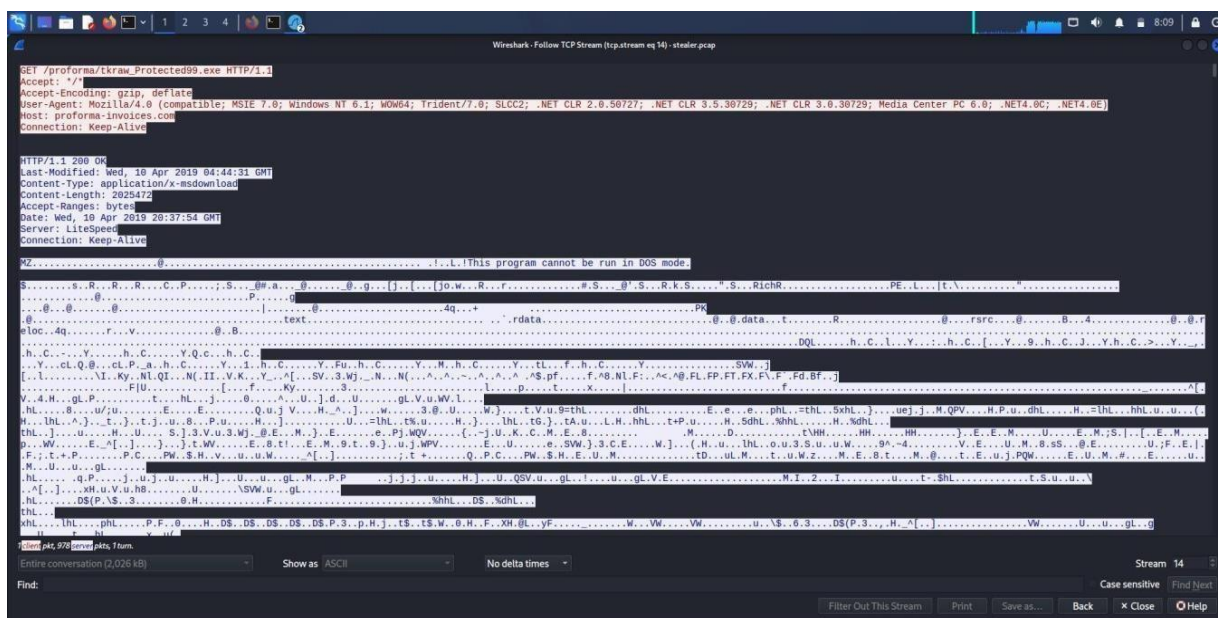


Step 8 - Following the TCP Stream to Inspect the Downloaded File

To get more details about the downloaded file, I selected the HTTP packet related to **tkraw_Protected99.exe** and used the **Follow TCP Stream** option in Wireshark. This opened the complete communication between the infected system and the web server. In the stream, I observed the HTTP request, server response, file information, and the binary content of the downloaded executable file. The stream also showed details such as the file name, host name, content type, and file size. By examining this TCP stream, I confirmed that the system downloaded a suspicious executable file from the remote server.

Key Findings:

- Downloaded file: **tkraw_Protected99.exe**
- Host: **proforma-invoices.com**
- HTTP response: **200 OK**
- Content-Type: **application/x-msdownload**
- Executable file signature: **MZ**

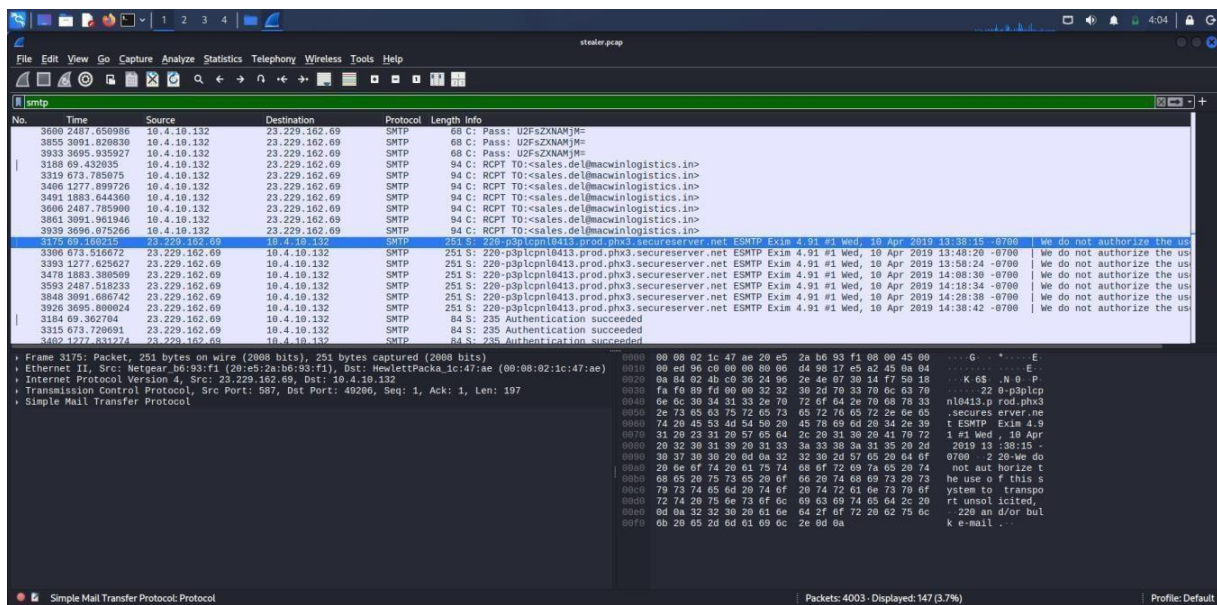


Step 9 - Analyzing SMTP Traffic

To check the email-related traffic in the PCAP file, I filtered the network traffic to display only SMTP packets.

In the Wireshark filter bar, I entered: **smtp**

After applying the filter, I examined the SMTP packets and observed communication between the internal system **10.4.10.132** and the external mail server **23.229.162.69**. I found SMTP authentication messages and email commands such as **RCPT TO**, which showed that email communication was taking place.



Step 10 - Following the TCP Stream to inspect SMTP Communication

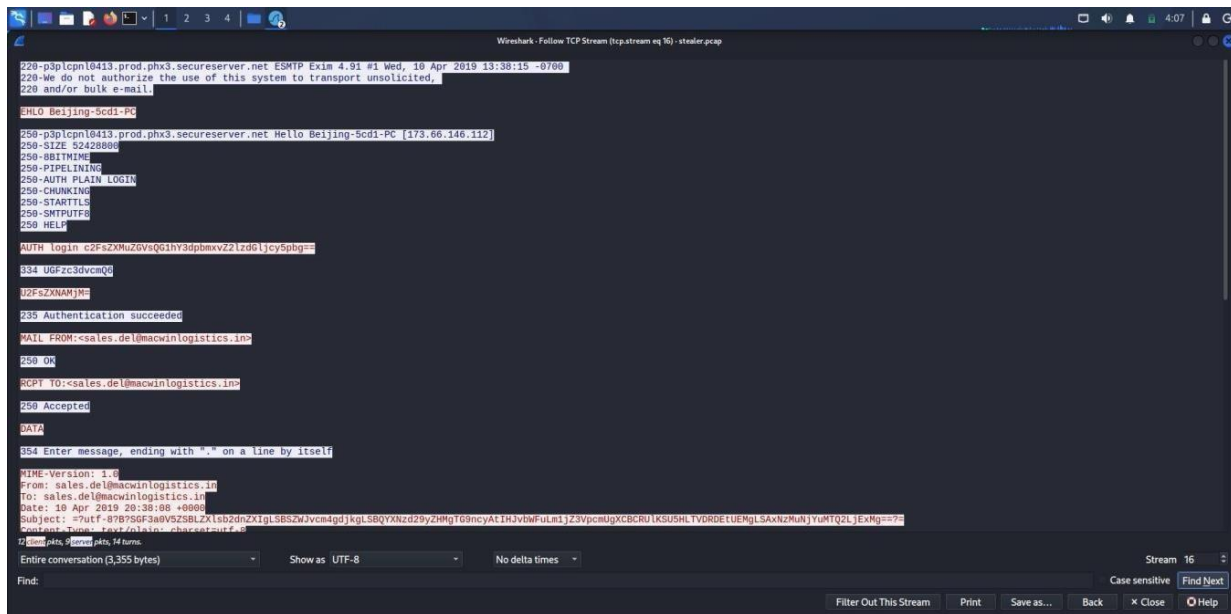
To get more details about the email communication, I selected an SMTP packet and used the **Follow TCP Stream** option in Wireshark. This opened the complete SMTP conversation between the computer and the mail server.

In the stream, I observed the **EHLO** command, SMTP authentication and the message **Authentication succeeded**. I also found the **MAIL FROM** and **RCPT TO** commands, which showed the sender and recipient involved in the email communication. The stream also displayed the email message information and other SMTP details.

By examining this TCP stream, I was able to understand the email communication more clearly and identify useful evidence from the network traffic.

Key Findings:

- **Protocol:** SMTP
- **SMTP server:** p3plcpnl0413.prod.phx3.secureserver.net
- **Client:** Beijing-5cd1-PC
- **Authentication:** Authentication succeeded
- **Sender:** sales.del@macwinlogistics.in
- **Recipient:** sales.del@macwinlogistics.in
- **Commands observed:** EHLO, AUTH, MAIL FROM, RCPT TO, DATA



Step 11 - Decoding the SMTP Credentials

To understand the encoded login details found in the SMTP stream, I used the **Base64 decoding** command in the Kali Linux terminal.

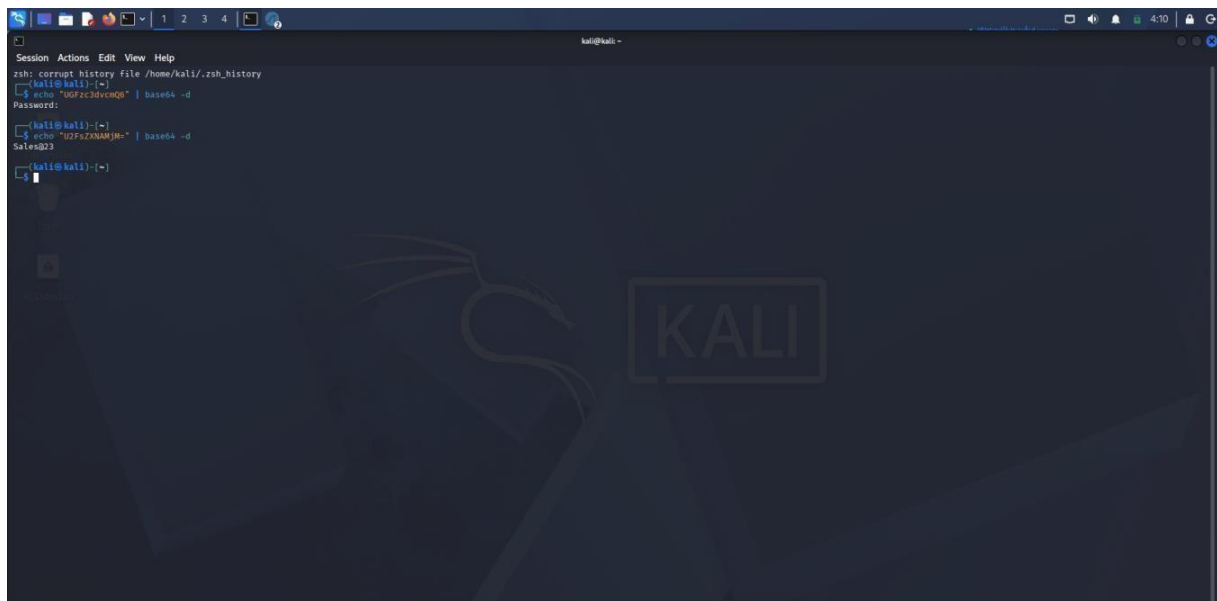
I used the following command: `echo "UGFzc3dvcmQ6" | base64 -d`

The output showed: **Password:**

I then decoded the other Base64 value using: `echo "U2FsZXNAMjM=" | base64 -d`

The output was: **Sales@23**

This helped me understand that the SMTP communication contained Base64-encoded credential information.

A screenshot of a Kali Linux terminal window. The terminal shows the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ echo "UGFzc3dvcmQ6" | base64 -d
Password:
kali@kali:~$ echo "U2FsZXNAMjM=" | base64 -d
Sales@23
kali@kali:~$
```

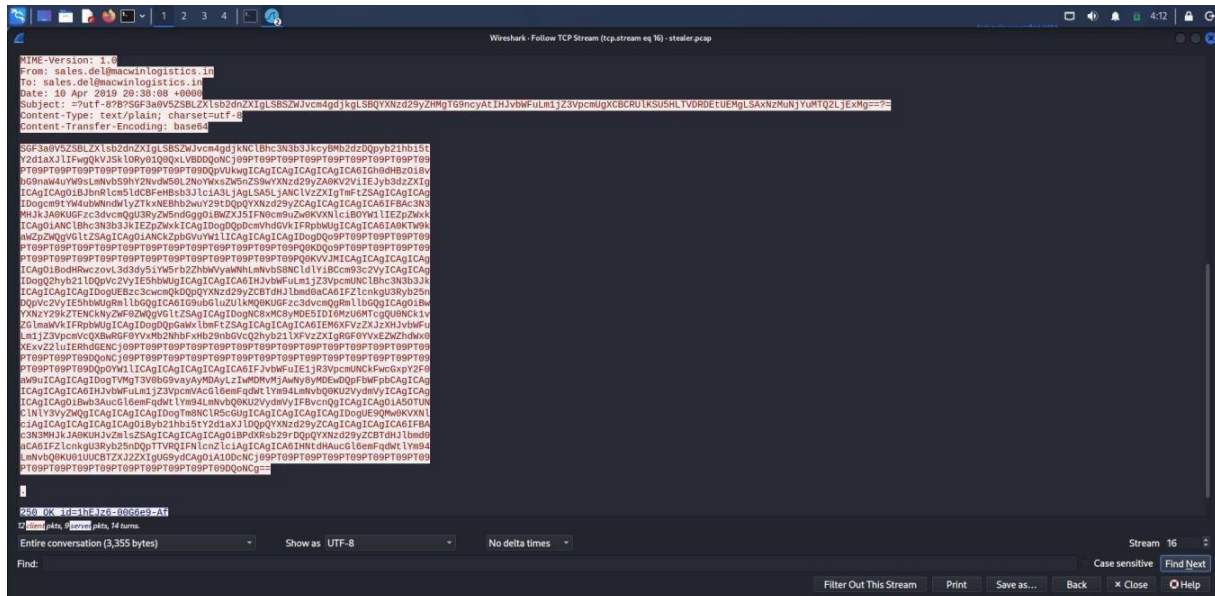
The terminal window has a dark blue background with a faint Kali Linux logo. The top of the window shows a menu bar with "Session", "Actions", "Edit", "View", and "Help". The status bar at the bottom shows the time as 4:10.

Step 12 - Identifying the Email Content

To examine the email communication in more detail, I continued reviewing the SMTP TCP stream in Wireshark. The stream showed email information such as the **From, To, Date, and Subject** fields. I also observed a long **Base64-encoded string** in the email content. The stream showed **Content-Transfer-Encoding: base64**, indicating that the email content was encoded before being transferred.

Key Findings:

- **From:** sales.del@macwinlogistics.in
- **To:** sales.del@macwinlogistics.in
- **Date:** 10 Apr 2019 20:38:08 +0000
- **Content-Type:** text/plain; charset=utf-8



Step 13 - Checking Protocol Statistics

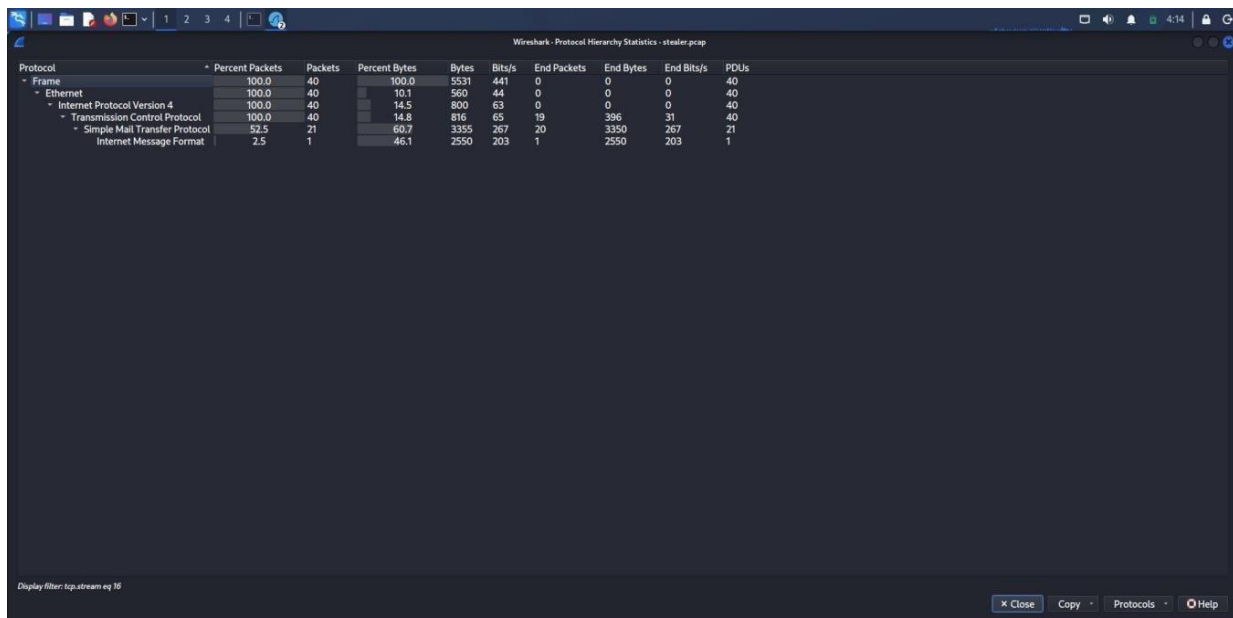
To get an overall view of the network traffic, I opened the **Protocol Hierarchy Statistics** in Wireshark.

The statistics showed the protocols present in the selected traffic. I observed **TCP**, **SMTP**, and **Internet Message Format (IMF)** in the results. This gave me a quick summary of the type and amount of traffic present in the investigated stream.

This helped me understand the protocol structure of the captured communication.

Key Findings:

- **TCP:** Transport protocol used for the communication
- **SMTP:** Email communication was present
- **Internet Message Format:** Email message data was observed
- **SMTP packets:** 21 packets
- **SMTP bytes:** 3,355 bytes



Wireshark - Protocol Hierarchy Statistics - stealer.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	40	100.0	5531	441	0	0	0	40
Ethernet	100.0	40	10.1	560	44	0	0	0	40
Internet Protocol Version 4	100.0	40	14.5	800	63	0	0	0	40
Transmission Control Protocol	100.0	40	14.8	816	65	19	396	31	40
Simple Mail Transfer Protocol	52.5	21	60.7	3355	267	20	3350	267	21
Internet Message Format	2.5	1	46.1	2550	203	1	2550	203	1

Display filter: tcp.stream eq 10

Close Copy Protocols Help

Step 14 - Solving the HawkEye Lab Questions

Using the evidence collected during the packet capture analysis, I answered all **24 questions** provided in the HawkEye challenge. The questions required analyzing different parts of the network traffic, including HTTP downloads, SMTP communication, credentials, malware-related information, and data exfiltration activity.

I checked the required information from the PCAP using Wireshark and submitted the answers on the CyberDefenders platform. After successfully answering all the questions, the lab showed **24/24 questions completed** and **100% completion**, confirming that the HawkEye challenge was completed successfully.

The screenshot shows the HawkEye Lab interface on the CyberDefenders platform. The header includes the lab title "HawkEye Lab" and a description: "Reconstruct a HawkEye Keylogger data exfiltration incident by analyzing network traffic with Wireshark and CyberChef, identifying IoCs and stolen credentials." Below this, there are tabs for "Category: Network Forensics", "Tactics: Initial Access, Execution, Stealth, Credential Access, Discovery, Collection, Command and Control, Exfiltration", and "Tools: Wireshark, Brim, Apackets, MaxMind Geo IP, VirusTotal". A difficulty level of "Medium" and a status of "Retired" are also shown. Action buttons include "Bookmark", "Join the Lab Squad", "Report an Issue", and "Share Achievement".

On the left sidebar, there is a "Download Lab Files" section with instructions to unzip the file using the password "cyberdefenders.org". Below this is a progress bar showing "24 / 24 Questions" and "100% Completed". There is also an "Official walkthrough" link and a "Featured Writeups" section with two entries: "Moshii" and "Sh3rlockH".

The main content area is titled "Scenario" and displays a list of questions. The first three questions are shown, all marked as "Solved":

- Q1** Solved : 6073
How many packets does the capture have?
Answer: 4003
- Q2** Solved : 5577
At what time was the first packet captured (UTC)?
Answer: 2019-04-10 20:37
- Q3** Solved : 5714
What is the duration of the capture?
Answer: 01:03:41

Each question has a "Hints" button and a "Submit" button.

5. Skills and Competencies Gained

- Developed a practical understanding of network traffic analysis and its use in cybersecurity investigations.
- Learned how to use **Wireshark** to filter and examine different types of network traffic.
- Improved my knowledge of **HTTP, TCP, DNS, and SMTP** traffic analysis.
- Learned how to identify suspicious files and activities from captured network traffic.
- Improved my skills in **PCAP analysis** by examining packets and following TCP streams.
- Learned how to identify and decode **encoded credentials** found in network traffic.
- Developed better understanding of **malware communication and data exfiltration** activities.

6. Feedback and Evidence

Feedback

This project gave me practical experience in investigating a network security incident using the HawkEye lab. By analyzing the PCAP file in Wireshark, I learned how to examine HTTP and SMTP traffic, follow TCP streams, identify suspicious file downloads and understand how credentials and email data were transmitted. The investigation also improved my understanding of how network evidence can be used to identify suspicious activities.

Evidence Collected

- Screenshots of the **HawkEye lab on CyberDefenders**.
- Wireshark screenshot showing the **PCAP file loaded**.
- Screenshot of the **HTTP traffic filter** and suspicious executable download.
- Screenshot showing the **HTTP TCP stream for tkraw_Protected99.exe**.
- Screenshots of the **SMTP traffic and SMTP TCP stream**.
- Screenshot showing the **Base64 decoding** of the captured data.
- Screenshot showing the **email content and headers**.
- Screenshot of the **Protocol Hierarchy Statistics**.
- Rule Screenshots showing 24/24 questions completed and 100% completion on CyberDefenders.

7. Challenges and Solutions

During the project, I faced some challenges while working with the HawkEye lab and analyzing the PCAP file in Wireshark. Since I was still learning network traffic analysis, understanding the large amount of captured traffic and finding the important packets required some time and practice. To overcome this, I used Wireshark filters such as `http` and `smtp` to focus on the required traffic.

Understanding the HTTP communication was also challenging at first because the packet capture contained a large amount of information. I used the **Follow TCP Stream** option to view the complete communication and identify the download of `tkraw_Protected99.exe`. This helped me understand how the file was transferred through the network.

The SMTP investigation was another challenging part because some of the information was encoded. I identified the Base64-encoded data and used the Kali Linux terminal to decode it. I also followed the SMTP TCP stream to understand the email communication and examine the available information. Despite these challenges, the investigation helped me improve my Wireshark skills and my ability to analyze network evidence.

8. Outcomes and Impact

This project gave me an opportunity to gain practical experience in investigating a malware-related network incident using Wireshark. During the HawkEye lab, I analyzed the provided PCAP file, examined HTTP and SMTP traffic, followed TCP streams, identified a suspicious executable download and investigated encoded information found in the communication.

Overall, the project was a useful learning experience that improved my understanding of **network traffic analysis, PCAP investigation, malware activity and digital forensics**. Completing the HawkEye lab and answering all 24 questions also helped me improve my problem-solving and investigation skills. The knowledge gained from this project will be useful for my future learning and work in cybersecurity.

9. Conclusion

This project helped me gain practical experience in analyzing network traffic and investigating a cybersecurity incident. By working on the HawkEye Lab, I learned how to use Wireshark to examine a PCAP file and identify suspicious activities.

During the investigation, I analyzed different types of network traffic, including HTTP and SMTP communication. I found a suspicious executable file, examined its download activity, and investigated SMTP traffic to understand the email communication and authentication process. I also decoded Base64-encoded information and collected important evidence from the network packets.

Overall, this was a useful hands-on learning experience. It helped me understand how a security analyst can investigate an attack by collecting and analyzing evidence from network traffic. The knowledge and skills gained from this project will be helpful for my future learning and career in cybersecurity.

HAWKEYE LAB

Red Team | Blue Team | Purple Team Report

A simple and detailed report based on the HawkEye network traffic investigation

Task: Analyze the HawkEye lab and prepare three reports showing the attack view, the defense and investigation view, and the handover between both teams.

Lab Environment: CyberDefenders HawkEye Lab

Main Analysis Tool: Wireshark

Analysis System: Kali Linux

Evidence: stealer.pcap packet
capture

1. Introduction

This report explains the HawkEye Lab from three cybersecurity perspectives: Red Team, Blue Team, and Purple Team. The packet capture provided in the lab was examined using Wireshark in Kali Linux. During the investigation, I reviewed HTTP and SMTP traffic, followed TCP streams, identified a suspicious executable download and examined encoded information.

The Red Team section reconstructs the possible attacker activity from the network evidence. The Blue Team section explains how the suspicious activity was found and investigated. The Purple Team section connects both views and shows how the findings can help improve detection and response.

2. Investigation Overview

- The HawkEye lab file was downloaded and prepared.
- The stealer.pcap file was opened in Wireshark.
- HTTP traffic was filtered to investigate suspicious web activity.
- SMTP traffic was filtered to investigate email communication.
- TCP streams were followed to view complete conversations.
- Base64-encoded information was decoded.
- Protocol statistics were reviewed.
- All 24 HawkEye lab questions were answered successfully.

3. Red Team Report

The Red Team report looks at the incident from the attacker point of view. In this lab, no real attack was performed. The purpose is to reconstruct the possible attack flow using the evidence found in the packet capture.

3.1 Attack Entry and Delivery

The lab scenario described suspicious activity after an employee opened a malicious email attachment. From an attacker point of view, an email can be used as an entry point to start or support a malware infection. The later network activity showed a suspicious executable file being requested and transferred.

3.2 Payload Download

While examining HTTP traffic, a GET request was found for **tkraw_Protected99.exe**. The communication involved **217.182.138.150**. The TCP stream showed an HTTP **200 OK** response, the host **proforma-invoices.com**, content type **application/x-msdownload**, and an **MZ** executable signature.

3.3 Communication After the Download

The packet capture also contained SMTP communication between the internal system **10.4.10.132** and the external mail server **23.229.162.69**. The SMTP TCP stream showed EHLO, AUTH, MAIL FROM, RCPT TO, and DATA commands, together with successful authentication.

3.4 Encoded Information

Base64-encoded information was present in the SMTP stream. The encoded values were decoded during the investigation, and the email content also showed Base64 transfer encoding. This indicates that information was prepared and transmitted in encoded form.

3.5 Red Team Findings

- The lab scenario involved a malicious email-related incident.
- **tkraw_Protected99.exe** was requested through HTTP.
- **217.182.138.150** and **proforma-invoices.com** were observed in the HTTP communication.
- HTTP 200 OK, **application/x-msdownload**, and an **MZ** signature were observed.
- SMTP traffic showed successful authentication and email commands.
- Encoded information and email content were present in the SMTP communication.

3.6 Red Team Conclusion

The Red Team view helped reconstruct a possible sequence: an email-related event, a suspicious executable download, and later SMTP communication containing authentication and encoded information. These findings provide the starting point for defensive investigation.

4. Blue Team Report

The Blue Team report explains how the suspicious activity was investigated from the defender's point of view. The main goal was to find important evidence, understand the communication, and collect useful indicators from the packet capture.

4.1 Preparing the Investigation

The **stealer.pcap** file was opened in Wireshark on Kali Linux. Because the capture contained different protocols, filters were used to focus on the most relevant traffic.

4.2 Investigating HTTP Traffic

The display filter **http** was used to show web traffic. A GET request for **tkraw_Protected99.exe** was identified. The related packet was then examined with Follow TCP Stream, which showed the complete request and response, file details, content type, and executable data.

4.3 Investigating SMTP Traffic

The display filter **smtp** was used to examine email-related traffic. The analysis showed communication between **10.4.10.132** and **23.229.162.69**. Following the SMTP TCP stream revealed EHLO, AUTH, MAIL FROM, RCPT TO, and DATA commands, as well as successful authentication.

4.4 Decoding Encoded Information

The SMTP stream contained Base64-encoded values. These values were decoded in the Kali Linux terminal to understand the information inside the communication. This showed that credential-related information and encoded email content were present.

4.5 Reviewing Email Content

The SMTP stream also showed email fields such as From, To, Date, and Subject. The sender and recipient shown in the captured stream were **sales.del@macwinlogistics.in**. The email content used Base64 transfer encoding and plain text message information.

4.6 Checking Protocol Statistics

Wireshark Protocol Hierarchy Statistics showed TCP, SMTP, and Internet Message Format traffic. In the selected SMTP communication, 21 SMTP packets and 3,355 bytes were shown.

4.7 Blue Team Evidence Collected

- Suspicious executable: **tkraw_Protected99.exe**
- HTTP server IP: **217.182.138.150**
- Host: **proforma-invoices.com**
- HTTP response: **200 OK**
- Content type: **application/x-msdownload**
- Executable signature: **MZ**
- SMTP communication between **10.4.10.132** and **23.229.162.69**

- SMTP commands: EHLO, AUTH, MAIL FROM, RCPT TO, DATA
- Successful SMTP authentication and Base64-encoded information
- Email headers and message content
- Protocol statistics showing TCP, SMTP, and Internet Message Format traffic

4.8 Blue Team Response Recommendations

The evidence should be preserved and the suspicious file, domain, and IP address should be recorded as investigation indicators. The affected system should be checked for signs of the downloaded executable. Outbound SMTP activity should be reviewed, and any exposed credentials should be changed and investigated.

Detection can be improved for suspicious executable downloads, unusual SMTP authentication, unexpected outbound mail activity, and suspicious encoded data sent outside the network.

5. Purple Team Report

The Purple Team combines the attacker and defender views. The purpose is to take the activity reconstructed from the Red Team side and use it to improve the Blue Team's monitoring and response.

5.1 Red Team to Blue Team Handover

The main sequence identified in the investigation was an email-related event, a suspicious executable download, and later SMTP communication with authentication and encoded information. The Blue Team can use this sequence to search for similar behavior in other network captures and security logs.

Attack activity	Evidence found	Defensive use
Suspicious executable download	tkraw_Protected99.exe, HTTP 200 OK, MZ signature	Alert on suspicious executable downloads and investigate the endpoint
Suspicious web infrastructure	217.182.138.150 and proforma-invoices.com	Search security logs for related connections and validate blocking
SMTP authentication	AUTH activity and authentication success	Review unusual outbound SMTP authentication
Email communication	MAIL FROM, RCPT TO, DATA, message headers	Monitor unusual mail flow and investigate unexpected communication
Encoded information	Base64-encoded values and content	Inspect suspicious encoded content and correlate with other events

5.2 Detection Improvements

The findings can be used to improve alerts for executable downloads through HTTP, especially when the content type and file signature indicate a Windows executable. Outbound SMTP traffic should also be reviewed when a workstation communicates with an unexpected mail server.

A single indicator may not prove an attack. The Purple Team approach is to combine multiple pieces of evidence. An email-related event, a suspicious executable download, and unusual outbound SMTP communication together provide a stronger reason for investigation.

5.3 Purple Team Learning Points

- Attack activity can be reconstructed from network evidence.
- Wireshark filters and TCP streams can reveal important details.
- Suspicious files, domains, and IP addresses can be used as investigation indicators.
- SMTP authentication and encoded data should be correlated with other events.
- Attack findings can be converted into monitoring ideas and investigation procedures.
- Sharing information between offensive and defensive teams improves security.

5.4 Purple Team Handover Summary

The HTTP and SMTP events should be connected as one investigation timeline. The analysis identified a suspicious executable download followed by SMTP communication containing authentication and encoded information. Connecting these observations gives defenders a clearer picture and supports better detection in the future.

6. Overall Findings

- A suspicious executable named tkraw_Protected99.exe was identified in HTTP traffic.
- 217.182.138.150 and proforma-invoices.com were observed in the HTTP communication.
- The HTTP response was 200 OK and the content type was application/x-msdownload.
- The transferred file showed an MZ executable signature.
- SMTP communication was observed between an internal system and an external mail server.
- The SMTP stream showed successful authentication and email commands.
- Base64-encoded information was identified and decoded.
- Email headers and encoded content were visible in the SMTP stream.
- Protocol statistics confirmed TCP, SMTP, and Internet Message Format traffic.
- All 24 HawkEye lab questions were completed successfully.

7. Skills Gained

- Using Wireshark to investigate PCAP files.
- Filtering HTTP and SMTP traffic.
- Following TCP streams.
- Identifying suspicious executable downloads.
- Understanding basic SMTP communication.
- Recognizing and decoding Base64 information.
- Collecting indicators and evidence from network traffic.
- Understanding Red Team, Blue Team, and Purple Team roles.

8. Conclusion

The HawkEye Lab was a useful practical exercise for understanding a malware-related network investigation. By analyzing the packet capture, I identified a suspicious executable download, examined SMTP communication, reviewed encoded information and collected evidence from different network protocols.

The Red Team view helped reconstruct the possible attack activity. The Blue Team view showed how the evidence was found and investigated. The Purple Team view connected both sides and showed how the findings can support monitoring and future detection.

Overall, this task improved my understanding of Wireshark, PCAP analysis, network traffic investigation, malware-related communication, and the relationship between Red Team, Blue Team and Purple Team activities.